1. Even q

Theorems **??**, **??** and **??** were stated for odd q. Here $q = 2^e$. Throughout, $\mathbb{F} = \mathrm{GF}(q)$, Tr is the absolute trace of $\mathbb{F}$, and we fix $\delta \in \mathbb{F}$ with $\mathrm{Tr}(\delta) = 1$, so that $t^2 + t + \delta$ is irreducible over $\mathbb{F}$ and the binary form $a^2 + ab + \delta b^2$ is anisotropic.

1.1. **Diameter 4.** For q even nothing new is needed: $\mathrm{Q}(4, q)$ has both ovoids and spreads [**?**, Ch. 3], and Corollary **??** already gives a Δ-regular structure with $2\Delta(q^2 + 1)$ elements for every $\Delta \geq q + 1$. In the language of doubling (Section **??**): if $\mathcal{O}$ is an ovoid and Σ a spread, every line of Σ contains exactly one point of $\mathcal{O}$, so $\mathcal{A}_0 = \mathcal{O} \cup \Sigma$ (one level) has $q^2 + 1$ components, each a flag, and satisfies **??** and (C_4). Doubling gives maximum degree $q + 2$, diameter 4 and $2(q + 1)(q^2 + 1) + 2(q^2 + 1) = 2(q^2 + 1)(q + 2)$ elements; the elements of $\mathcal{A}_0$ and their copies have degree $q + 1$, and adding the anti-flags (p, L') and (p', L) for each flag (p, L) of $\mathcal{A}_0$ makes the structure $(q+2)$-regular. This is the structure of Corollary **??** with $w = \mathbf{1}_{\mathcal{O}}$, $\mathcal{M} = \Sigma$, made regular. Every element outside $\mathcal{A}_0$ is incident with exactly one element of $\mathcal{A}_0$, so the counting bound of Proposition **??** holds with equality for $q^2 + 1$ components.

At $\Delta = q+2$ this beats the value $2q^2(q+2)$ of Theorem **??** by $2(q+2)$, so for even q the quadrangle $\mathrm{AS}(q)$ is not needed. For $q = 2, 4, 8, 16$ these graphs (and the ones with $j = 2$ copies of $\mathcal{O}$ and of Σ) were built and checked; see Table 1. They do not improve the table: when $q \in \{2, 4, 8, 16\}$, $q + 1$ is a prime power, so $\Delta = q + 2$ is a Moore degree, and at $\Delta = q + 3$ the odd prime power $q + 1$ gives more. The first degrees at which an even q is the best choice in Corollary **??** are $\Delta = 34, 35, 36, 37$ ($q = 32$; for example $n_b(34, 4) \geq 2 \cdot 1025 \cdot 34 = 69700$, against 65416 from $q = 31$), outside the range of the table.

1.2. **Diameter 6: the frame in characteristic 2.** The split Cayley hexagon $\mathrm{H}(q)$ is defined for all q by the same quadric and the same six Tits equations; in characteristic 2 the signs disappear and the quadric $\mathrm{Q}(6, q)\colon X_0X_4 + X_1X_5 + X_2X_6 + X_3^2 = 0$ has a nucleus, the point e_3. The hyperplanes $X_0 + cX_4 + \cdots = 0$ of Section **??** contain e_3 and so meet $\mathrm{Q}(6, q)$ in a cone; instead we take, for $z = (\alpha, \beta) \in \mathbb{F}^2$,

$$\Pi_z\colon\ X_3 = X_0 + \delta X_4 + \alpha X_2 + \beta X_5, \qquad \Pi = \Pi_{(0,0)},$$

and S_z the set of lines of $\mathrm{H}(q)$ in Π_z. Again $v = \langle e_1, e_6 \rangle \subseteq \Pi_z$ for all z.

Lemma 1.1. *Lemma* **??** *holds verbatim for even q with these Π_z.*

Proof. (a) On Π_z,

$$X_0X_4 + X_1X_5 + X_2X_6 + X_3^2 = \left(X_0^2 + X_0X_4 + \delta^2X_4^2\right) + (X_1 + \beta^2X_5)X_5 + X_2(X_6 + \alpha^2X_2).$$

The first term is anisotropic since $\mathrm{Tr}(\delta^2) = \mathrm{Tr}(\delta) = 1$, and Π_z misses the nucleus e_3, so $\Pi_z \cap \mathrm{Q}(6, q)$ is a non-degenerate quadric of Witt index 2, an elliptic quadric $\mathrm{Q}^-(5, q)$. (b), (c) and (e) are proved as before. (d) The Tits equations give $X_2 = X_5 = 0$ exactly as before (only signs change); the cone $X_0X_4 = X_3^2$ meets Π_z where $X_0^2 + X_0X_4 + \delta^2X_4^2 = 0$, that is $X_0 = X_4 = 0$, hence $X_3 = 0$, on v. $\square$

So Theorem **??** (Delorme's doubling with whiskers) holds for even q without change. Free points are defined as before, there are $q^3(q^2 - 1)$ of them, and an exact whisker set has $q^2(q^2 - 1)$ lines.

1.3. **An exact whisker set for even q.** For a point X of $\mathrm{H}(q)$ in Π, not on v, put

$$\nu(X) = \frac{X_2X_3 + X_2X_4 + X_4X_5}{\delta X_2^2 + X_2X_5 + X_5^2}.$$

The denominator is anisotropic, and vanishes only if $X_2 = X_5 = 0$, that is (Lemma 1.1) only on v. For $c_0, c_1 \in \mathbb{F}$ let

$$G(a, b) = (c_0a + c_1b)(a^2 + ab + \delta b^2) + b^3, \qquad F(z) = \frac{G(\alpha, \beta)}{\alpha^2 + \alpha\beta + \delta\beta^2} \quad (z = (\alpha, \beta) \neq 0).$$

Lemma 1.2. *There are $c_0, c_1 \in \mathbb{F}$ such that G has no zero on $\mathbb{F}^2 \setminus \{0\}$ if and only if $q \geq 4$.*

Proof. $G(a,0) = c_0 a^3$, so $c_0 \neq 0$ is needed. For $b \neq 0$ and $t = a/b$, $G(a,b) = 0$ if and only if $c_0 t + c_1 = r(t) := 1/(t^2 + t + \delta)$. For fixed t this is a line in the (c_0, c_1)-plane with $q - 1$ points having $c_0 \neq 0$. If $q \geq 4$, take $\omega \notin \{0, 1\}$; then $r(\omega) \neq r(0)$, so the lines for $t = 0$ and $t = \omega$ meet at $c_0 = (r(0) + r(\omega))/\omega \neq 0$, and the q lines cover fewer than $q(q-1)$ points with $c_0 \neq 0$. If $q = 2$, then $r(0) = r(1) = 1$ and the lines $c_1 = 1$, $c_0 + c_1 = 1$ contain both points with $c_0 = 1$. □

We use coordinates $x = (x_0, x_1, x_2, x_4, x_5, x_6)$ on Π_z, with $X_3 = x_0 + \delta x_4 + \alpha x_2 + \beta x_5$, and define the linear map J_z of Π_z by

$$J_z x = \big((\delta+1)x_0 + \delta\alpha x_2 + \delta^2 x_4 + (\alpha+\delta\beta)x_5,\quad \beta x_0 + x_1 + \alpha\beta x_2 + (\alpha+\delta\beta)x_4 + \beta^2 x_5 + x_6,$$
$$x_2 + x_5,\quad x_0 + (\alpha+\beta)x_2 + \delta x_4 + \beta x_5,\quad \delta x_2,\quad (\alpha+\beta)x_0 + \delta x_1 + \alpha^2 x_2 + \delta\alpha x_4 + \alpha\beta x_5 \big).$$

Lemma 1.3 (Complex structure, characteristic 2)**.** *For every $z \in \mathbb{F}^2$, with Q the quadratic form and B its polar form:*

(a) *$J_z^2 + J_z + \delta = 0$, $Q(J_z x) = \delta Q(x)$ and $B(x, J_z x) = Q(x)$.*
(b) *The six Tits differences of the Grassmann coordinates of $x \wedge J_z x$ equal $Q(x)$ times $1, 0, 0, \delta, \alpha, \beta$.*
(c) *For every point x of $\mathrm{Q}(6,q) \cap \Pi_z$ the line $\langle x, J_z x\rangle$ is the line of S_z on x.*
(d) *With* $\mathrm{num} = X_2X_3 + X_2X_4 + X_4X_5$ *and* $\mathrm{den} = \delta X_2^2 + X_2X_5 + X_5^2$ *on Π:* $\mathrm{num}(sx + tJ_0 x) = (s^2 + st + \delta t^2)\,\mathrm{num}(x)$*, and likewise for* den*. So ν is constant on every line of $S \setminus \{v\}$.*

Proof. (a), (b), (d) are polynomial identities over $\mathbb{F}_2[\delta, \alpha, \beta, x]$, checked by direct expansion (and by computer algebra, `research/evenq/proof_J.py`). (c) follows from (a), (b) and Lemma 1.1(b) as in Lemma **??**: J_z has no eigenvector because $t^2 + t + \delta$ is irreducible, and on a singular x the line $\langle x, J_z x\rangle$ is totally singular with Grassmann coordinates satisfying Tits' equations. In (d), $s^2 + st + \delta t^2 \neq 0$ for $(s,t) \neq 0$. □

Let p be a free point. Then $(p_2, p_5) \neq (0,0)$, and we normalize $p_0 + p_3 + \delta p_4 = 1$. Now $p \in \Pi_z$ if and only if z lies on the affine line $A_p\colon \alpha p_2 + \beta p_5 = 1$, which has direction (p_5, p_2). For $z \in A_p$ let $X(z)$ be the point of Π on the line of S_z through p.

Lemma 1.4 (Key identity, characteristic 2)**.** *For $z = (\alpha, \beta) \in A_p$,*

$$\Phi_p(z) := \nu(X(z)) + \frac{\beta^3}{\alpha^2 + \alpha\beta + \delta\beta^2} = \frac{p_2p_3 + p_2p_4 + p_4p_5 + \beta p_2^2}{\delta p_2^2 + p_2p_5 + p_5^2}.$$

So Φ_p is affine on A_p, with increment $p_2^3/(\delta p_2^2 + p_2p_5 + p_5^2)$ along (p_5, p_2).

Proof. Put $\Theta = \delta p_2^2 + p_2p_5 + p_5^2 \neq 0$ and $\lambda(y) = y_0 + y_3 + \delta y_4$, so that $\Pi = \{\lambda = 0\}$ and $\lambda = \alpha y_2 + \beta y_5$ on Π_z. Then $\lambda(p) = 1$ and $\gamma := \lambda(J_z p) = \alpha(p_2 + p_5) + \delta\beta p_2$. By Lemma 1.3(c) the line of S_z on p is $\langle p, J_z p\rangle$, so $X(z) = \gamma p + J_z p$ (then $\lambda(X) = 2\gamma = 0$). In particular $X_2 = (\gamma+1)p_2 + p_5$ and $X_5 = \delta p_2 + \gamma p_5$. Direct expansion gives

$$\mathrm{den}(X) = (\gamma^2 + \gamma + \delta)\,\Theta, \qquad \mathrm{num}(X) = (\gamma^2 + \gamma + \delta)(p_2p_3 + p_2p_4 + p_4p_5) + \beta\big((\gamma+\delta+1)p_2^2 + p_5^2\big),$$

and $\gamma^2 + \gamma + \delta \neq 0$. Solving $\alpha p_2 + \beta p_5 = 1$, $\alpha(p_2 + p_5) + \delta\beta p_2 = \gamma$ (determinant Θ) gives $\alpha = (\delta p_2 + \gamma p_5)/\Theta$, $\beta = ((\gamma+1)p_2 + p_5)/\Theta$, hence $\alpha^2 + \alpha\beta + \delta\beta^2 = (\gamma^2 + \gamma + \delta)/\Theta$ and

$$\frac{\beta^3}{\alpha^2 + \alpha\beta + \delta\beta^2} = \frac{\beta\big((\gamma+1)p_2 + p_5\big)^2}{(\gamma^2 + \gamma + \delta)\Theta}.$$

The β-terms add up to $\beta\big((\gamma+\delta+1) + (\gamma+1)^2\big)p_2^2/((\gamma^2 + \gamma + \delta)\Theta) = \beta p_2^2/\Theta$. Along (p_5, p_2), β changes by p_2. □

Theorem 1.5. *Let $q \geq 4$ be even and c_0, c_1 as in Lemma 1.2. Then*

$$W = \{\, m \in S_z \setminus \{v\} \;:\; z \in \mathbb{F}^2 \setminus \{0\},\ \nu(m \cap \Pi) = F(z) \,\}$$

is an exact whisker set, so $|W| = q^2(q^2-1)$. For $q = 2$ an exact whisker set (12 lines) exists as well (found by computer); the formula gives none, since no (c_0, c_1) is good.

Proof. As in Theorem **??**: the lines of $\bigcup_{z \neq 0} S_z \setminus \{v\}$ on a free point p are the q distinct lines $m_p(z)$, $z \in A_p$, and $m_p(z) \in W$ if and only if $\Phi_p(z) + c_0\alpha + c_1\beta = 0$, because $F(z) = c_0\alpha + c_1\beta + \beta^3/(\alpha^2 + \alpha\beta + \delta\beta^2)$. By Lemma 1.4 the left side is affine on A_p with increment

$$\frac{p_2^3}{\Theta} + c_0 p_5 + c_1 p_2 = \frac{G(p_5, p_2)}{\Theta} \neq 0$$

along (p_5, p_2), so it vanishes at exactly one point of A_p. □

Corollary 1.6. *Corollary **??** and Theorem **??** hold for every prime power q, odd or even: for $m \geq 2$,*

$$n_b(q+m, 6) \geq 2(q+1)(q^4+q^2+1) + (m-1)\big(2q^4+2q^3+2q+2+m(q^3+q+2)+m^2\big),$$

and there is a $(q+m)$-regular bipartite graph of diameter 6 on $2(q+1)\big(q^4+q^2+1+(m-1)(q^3+m+1)\big)$ vertices.

Proof. Theorem **??** with Lemma 1.1 and Theorem 1.5; the proof of Theorem **??** uses only that W is exact. □

As for $D = 4$, this does not change the table: for $q = 2, 4, 8, 16$ the degree $q+2$ is a Moore degree ($\mathrm{H}(q+1)$ exists), and at $\Delta = q+3$ the odd $q+1$ gives more. Even q is the best choice first at $\Delta = 34, \ldots, 37$ ($q = 32$; e.g. $n_b(34, 6) \geq 71502028$ against 63159250 from $q = 31$, $m = 3$).

Remark 1.7 (What changes in characteristic 2). The anisotropic form $a^2 + cb^2$ ($-c$ a non-square) becomes $a^2 + ab + \delta b^2$ ($\mathrm{Tr}\delta = 1$); $J^2 = -c$ becomes $J^2 + J + \delta = 0$; the hyperplanes Π_z must avoid the nucleus; ν gains the term X_2X_4; and Lemma **??** needs $q \geq 4$. Equivalently, with $K = \mathbb{F}[\omega]$, $\omega^2 = \omega + \delta$: $\nu(X) = \mathrm{Tr}_{K/\mathbb{F}}\big(\omega\,(X_3 + X_4\omega)/(\delta X_2 + X_5\omega)\big)$.

D	q	Δ	construction	order	check
4	2, 4, 8, 16	$q+2$	ovoid + spread, regular	40, 204, 1300, 9252	all VALID
4	2, 4, 8, 16	$q+3$	2 ovoids + 2 spreads, regular	50, 238, 1430, 9766	all VALID
6	2, 4, 8	$q+2$	Thm. **??** + exact W	208, 3524, 85180	all VALID
6	2, 4, 8	$q+3$	same, $m = 3$	324, 4468, 96516	all VALID
6	2, 4, 8	$q+2$	regular (Thm. **??**)	192, 3400, 84168	all VALID
6	2, 4, 8	$q+3$	regular, $m = 3$	270, 4090, 93474	all VALID
6	16	18	Thm. **??** + exact W	2384492	PENDING

TABLE 1. Even q, computer checks (bipartite, maximum degree, exact diameter by BFS from every vertex). For $D = 6$, $q \geq 4$ uses the whiskers of Theorem 1.5; $q = 2$ uses an ILP solution. Exactness of W was also checked directly for $q = 4$ ($\delta = 2, 3$), $q = 8$ ($\delta = 1, 3, 5, 7$) and $q = 16$ ($\delta = 8, 9$; 65280 whiskers). All values are below the Moore graphs or table values at the same Δ.